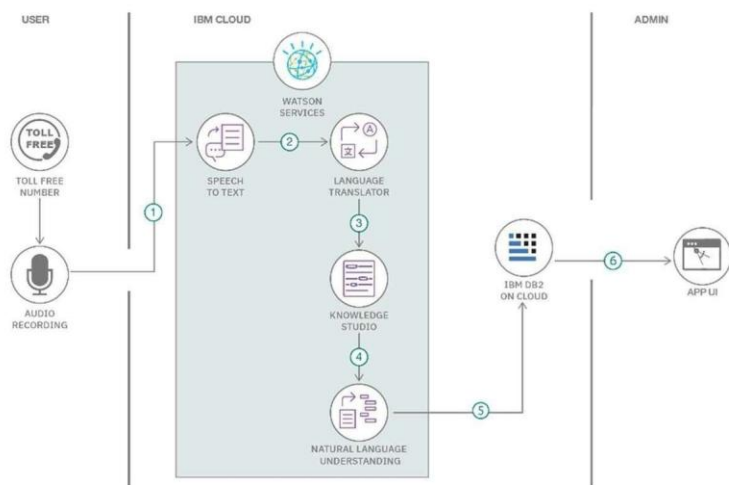


Project Design Phase-II  
Technology Stack (Architecture & Stack)

|               |                                                                                         |
|---------------|-----------------------------------------------------------------------------------------|
| Date          | 31 January 2026                                                                         |
| Team ID       | LTVIP2026TMIDS24317                                                                     |
| Project Name  | I Revolution_ A Data-driven Exploration of Apple's iPhone Impact in India using tableau |
| Maximum Marks | 4 Marks                                                                                 |

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table 1:

| Component     | Tool / Technology               | Purpose                                      |
|---------------|---------------------------------|----------------------------------------------|
| Data Source   | CSV, JSON files                 | Raw smartphone sales and specifications data |
| Visualization | Tableau Desktop                 | Creating interactive dashboards and stories  |
| Storage       | Google Drive / Local System     | Storing raw and processed datasets           |
| Collaboration | Google Docs, Slack              | Team communication and report writing        |
| Deployment    | Tableau Public / Tableau Server | Dashboard sharing and stakeholder access     |

Table 2: Application Characteristics

| S.No | Characteristics          | Description                                                                                                                                                                       | Technology                                    |
|------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Open-Source Frameworks   | The application uses open datasets and visualization tools to analyze iPhone sales and market trends in India. Tableau supports integration with open data formats for analytics. | Tableau Desktop, CSV/JSON data formats        |
| 2    | Security Implementations | The system ensures secure access to dashboards through Tableau Public/Server permissions. Data files are stored securely and access is controlled at the platform level.          | Tableau authentication, Access control, HTTPS |
| 3    | Scalable Architecture    | The architecture supports scalability by handling large datasets and multiple visualizations efficiently through Tableau's in-memory engine.                                      | Tableau Hyper Engine                          |
| 4    | Availability             | The dashboards are available online through Tableau Public/Server, enabling access anytime through a web browser without local installation.                                      | Tableau Public / Tableau Server               |

| S.No | Characteristics | Description                                                                                                                                    | Technology                            |
|------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 5    | Performance     | Performance is optimized using data extracts, filters, and efficient visual rendering to ensure fast dashboard loading and smooth interaction. | Tableau Extracts (.hyper),<br>Caching |